In [1]:
```
x = 10
y = 20
total = x + y
print(total)
```

```
30
```

Line Magic Command

%ls

In [3]:
```
%ls
```

```
 Volume in drive D is New Volume
 Volume Serial Number is B20E-3C36

 Directory of D:\AY-2526\Python for Data Science-SRM

17-06-2025  09:22    <DIR>          .
17-06-2025  09:21    <DIR>          ..
17-06-2025  09:22    <DIR>          .ipynb_checkpoints
13-06-2025  10:47    <DIR>          Encrypted PPT
31-05-2025  12:51    <DIR>          File-Python for Data Science
14-06-2025  10:33    <DIR>          Jupyter Files for Labs
31-05-2025  11:50    <DIR>          Lab File Pandas
12-06-2024  07:58            11,836 Lab planning-old.xlsx
04-05-2024  10:06            11,670 Lesson Planning-old.xlsx
11-06-2025  11:05           471,904 MCA3_RollNo_Batch_w_e_f_10_06_25.pdf
11-06-2025  11:08            12,317 MCA3_RollNo_Batch_w_e_f_10_06_25.xlsx
02-06-2025  12:07            12,099 New Microsoft Excel Worksheet.xlsx
14-06-2025  23:49                21 p.py
14-06-2025  23:50    <DIR>          PDS Lecture Notes
14-06-2025  10:47    <DIR>          Python for DS-Lecture jupyter Files-old
13-05-2025  13:31    <DIR>          Python for DS-MCQ
11-06-2025  09:13    <DIR>          Python for DS-PPT(SRM)
16-06-2025  08:47    <DIR>          Student Details
28-03-2025  10:43           301,156 Subject_Syllabus2305CS303.pdf
11-06-2025  19:08            12,508 TTM_LessonPlanningFormat_Sample_WithoutDateAn
dICTTools.xlsx
17-06-2025  09:22               337 Untitled.ipynb
               9 File(s)        833,848 bytes
              12 Dir(s)  114,285,527,040 bytes free
```

%pwd

In [5]:
```
%pwd
```

Out[5]:
```
'D:\\AY-2526\\Python for Data Science-SRM'
```

%cd

In [7]:
```
%cd ..
```

```
D:\AY-2526
```

```
C:\Users\mania\anaconda3\Lib\site-packages\IPython\core\magics\osm.py:417: UserWa
rning: This is now an optional IPython functionality, setting dhist requires you
to install the `pickleshare` library.
  self.shell.db['dhist'] = compress_dhist(dhist)[-100:]
```

```
In [9]: %pwd
```

```
Out[9]: 'D:\\AY-2526'
```

```
In [11]: %cd "Python for Data Science-SRM"
```

```
D:\AY-2526\Python for Data Science-SRM
```

```
C:\Users\mania\anaconda3\Lib\site-packages\IPython\core\magics\osm.py:417: UserWa
rning: This is now an optional IPython functionality, setting dhist requires you
to install the `pickleshare` library.
  self.shell.db['dhist'] = compress_dhist(dhist)[-100:]
```

%time

```
In [21]: %time total = sum(range(10000000))
```

```
CPU times: total: 484 ms
Wall time: 515 ms
```

%who

```
In [23]: %who
```

```
dataframe_columns        dataframe_hash   dtypes_str      get_dataframes   getpass
hashlib   import_pandas_safely    is_data_frame   json
total     x        y
```

%whos

```
In [25]: %whos
```

```
Variable                Type        Data/Info
--------------------------------------------
dataframe_columns       function    <function dataframe_colum<...>ns at 0x000002D7
789D42C0>
dataframe_hash          function    <function dataframe_hash at 0x000002D7789D6200
>
dtypes_str              function    <function dtypes_str at 0x000002D7789D7420>
get_dataframes          function    <function get_dataframes at 0x000002D7789D7880
>
getpass                 module      <module 'getpass' from 'C<...>conda3\\Lib\\get
pass.py'>
hashlib                 module      <module 'hashlib' from 'C<...>conda3\\Lib\\has
hlib.py'>
import_pandas_safely    function    <function import_pandas_s<...>ly at 0x000002D7
77B07600>
is_data_frame           function    <function is_data_frame at 0x000002D7789D6CA0>
json                    module      <module 'json' from 'C:\\<...>\Lib\\json\\__in
it__.py'>
total                   int         49999995000000
x                       int         10
y                       int         20
```

%reset

```
In [27]: %reset
```

```
In [29]: %who
```

```
dataframe_columns        dataframe_hash   dtypes_str       get_dataframes   getpass
hashlib   import_pandas_safely    is_data_frame    json
```

```
In [31]: x
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[31], line 1
----> 1 x

NameError: name 'x' is not defined
```

%run filename.py

```
In [33]: %run p.py
```

```
This is from Python file.
```

%matplotlib inline

```
In [35]: %matplotlib inline
```

%history

```
In [41]: %history -n 7
```

```
   7: %cd ..
```

%load filename.py

```
In [43]: # %load p.py
         print("This is from Python file.")
```

Cell Magic Commands

%%time

```
In [47]: %%time
         total = 0
         for i in range(1_00_000):
             total += i
         print("Total is",total)
```

```
Total is 4999950000
CPU times: total: 15.6 ms
Wall time: 13.4 ms
```

%%timeit

In [49]:
```
%%timeit
total = 0
for i in range(1_00_000):
    total += i
print("Total is",total)
```

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14.3 ms ± 257 µs per loop (mean ± std. dev. of 7 runs, 100 loops each)
```

%%writefile filename.py

In [51]:

```
%%writefile file1.py
total = 0
for i in range(1_00_000):
    total += i
print("Total is",total)
```

```
Writing file1.py
```